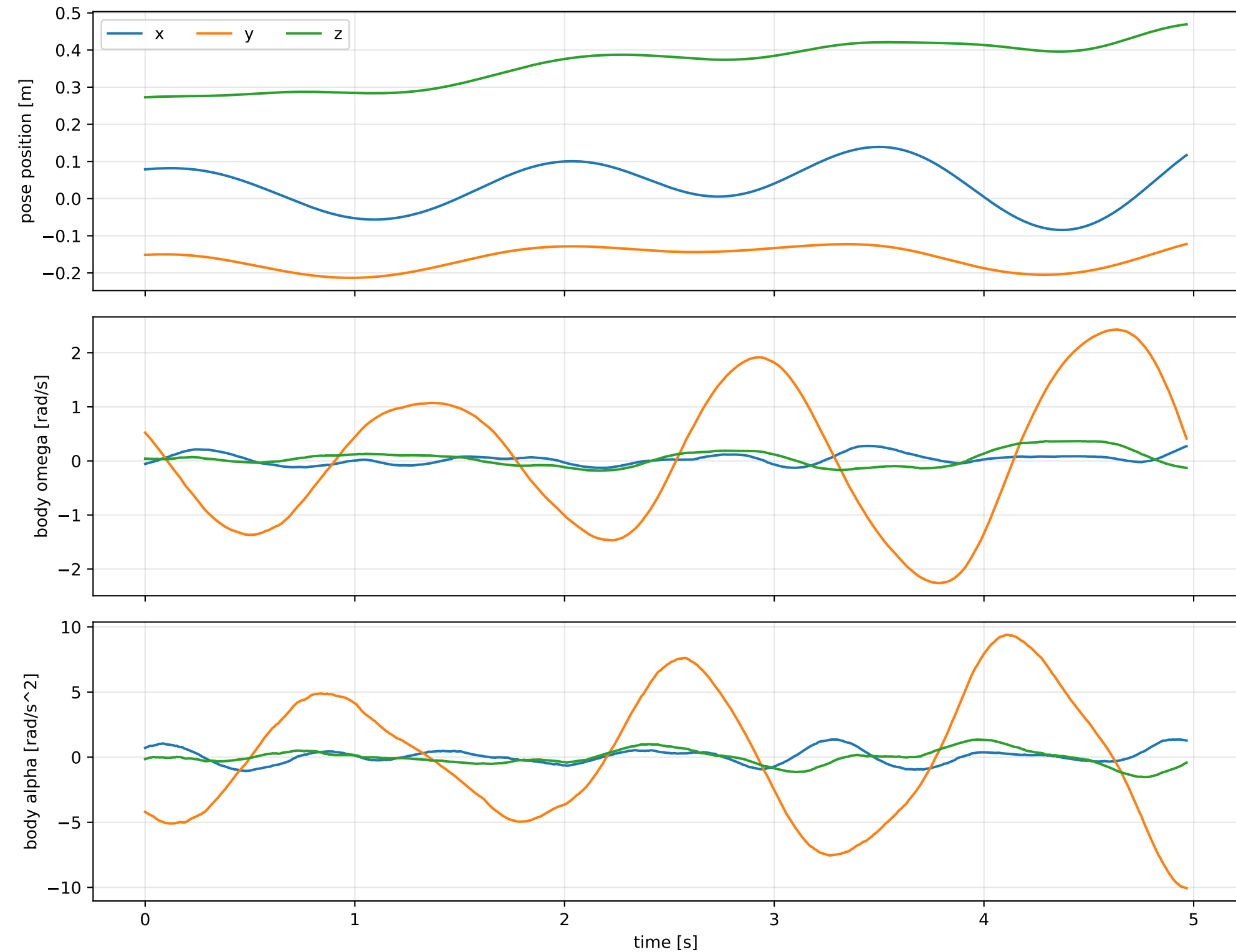
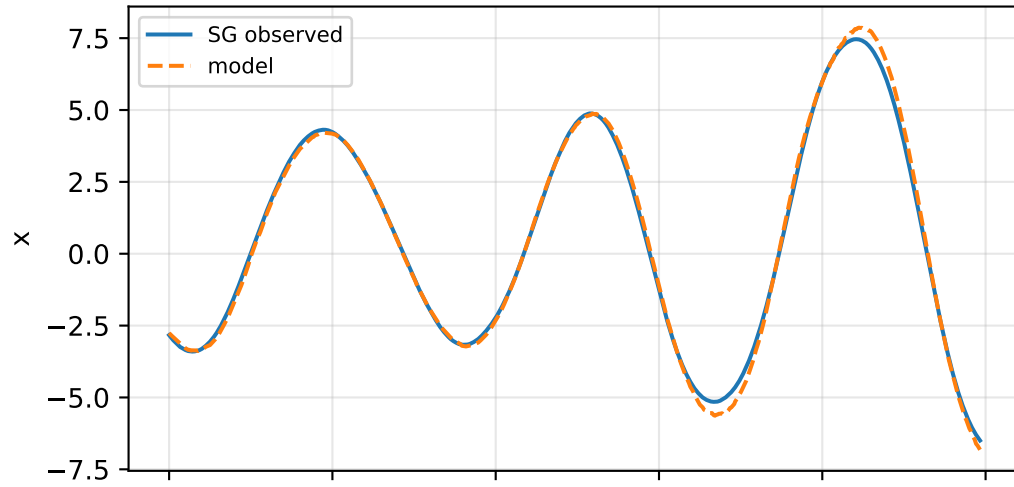


lag_legacy_smooth_schedule: SG trajectory and derived motion

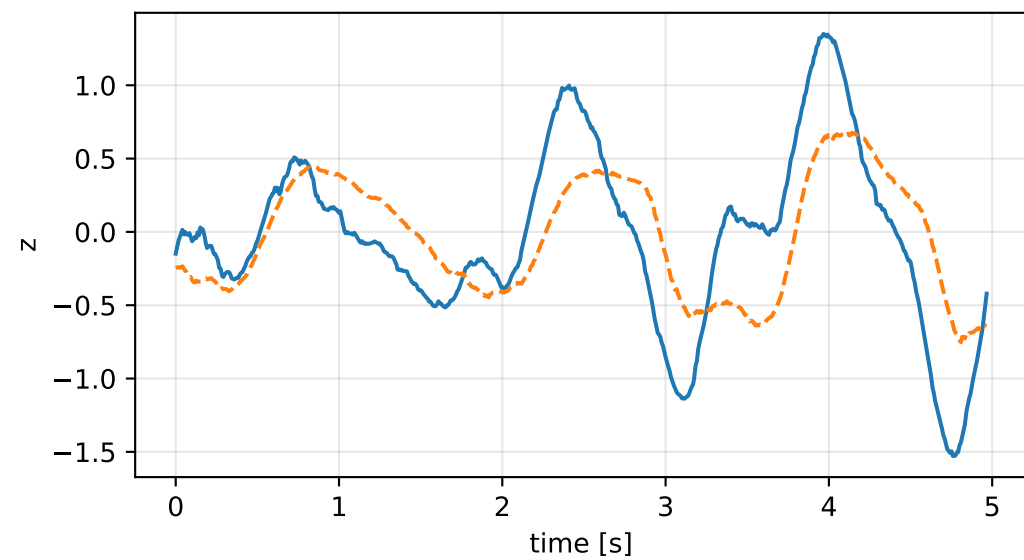
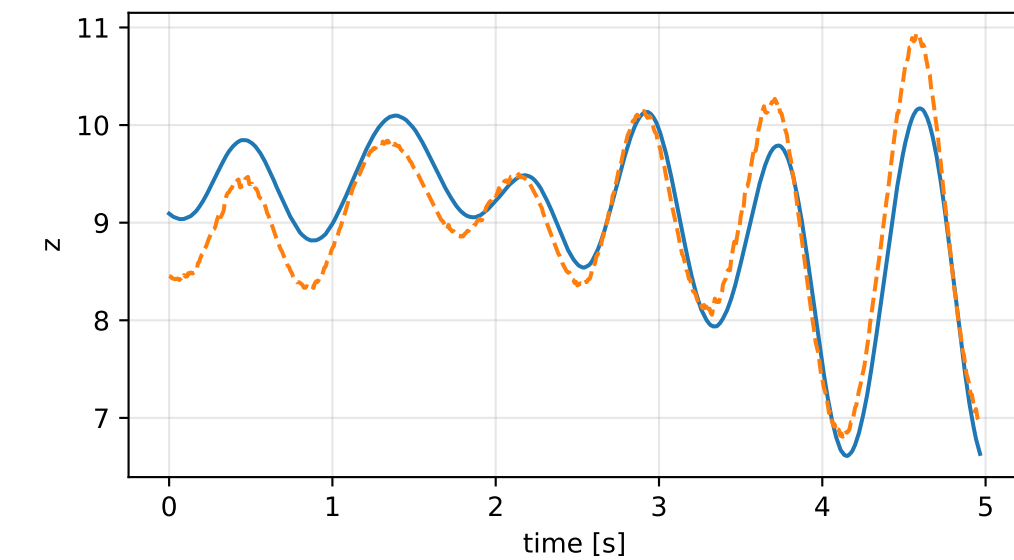
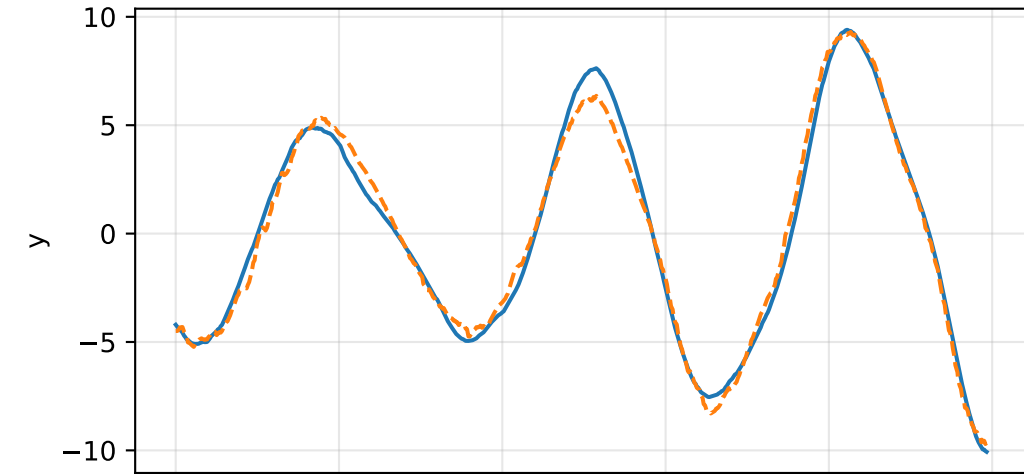
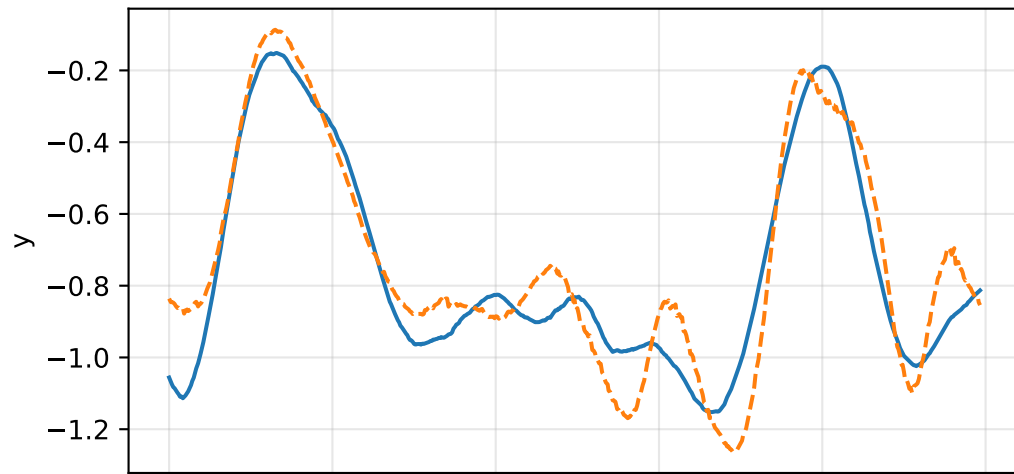
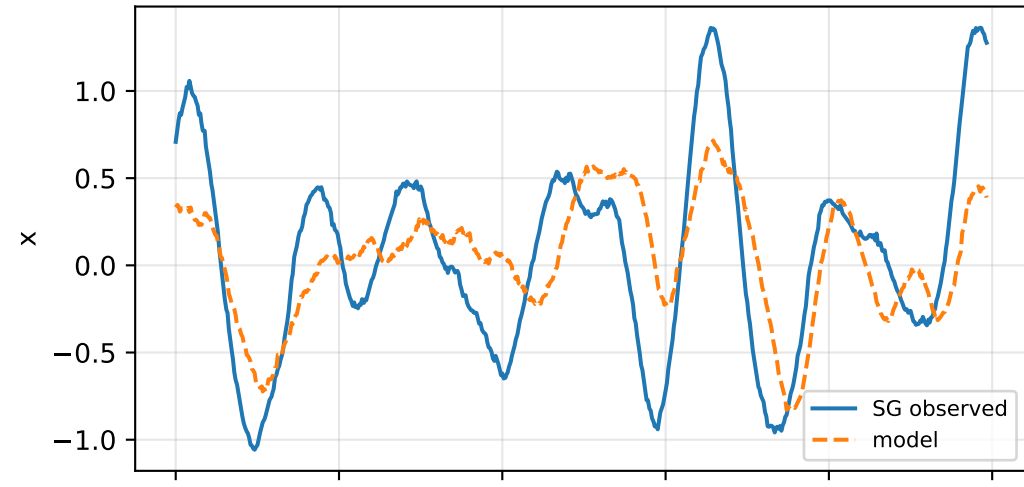


lag_legacy_smooth_schedule: acceleration residual objective

specific acceleration [m/s^2]

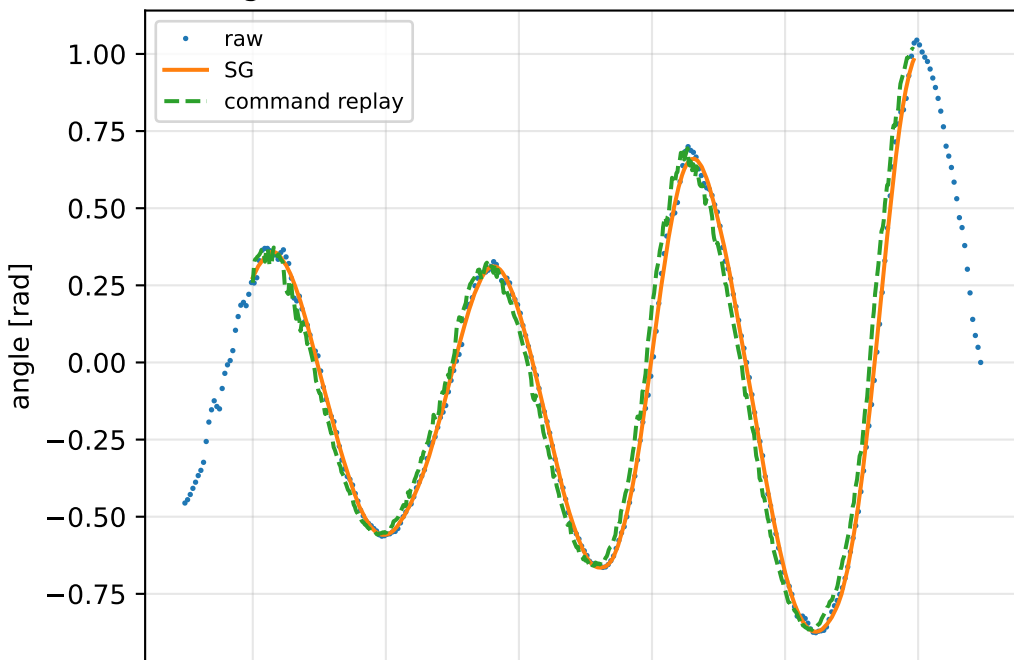


angular acceleration [rad/s^2]

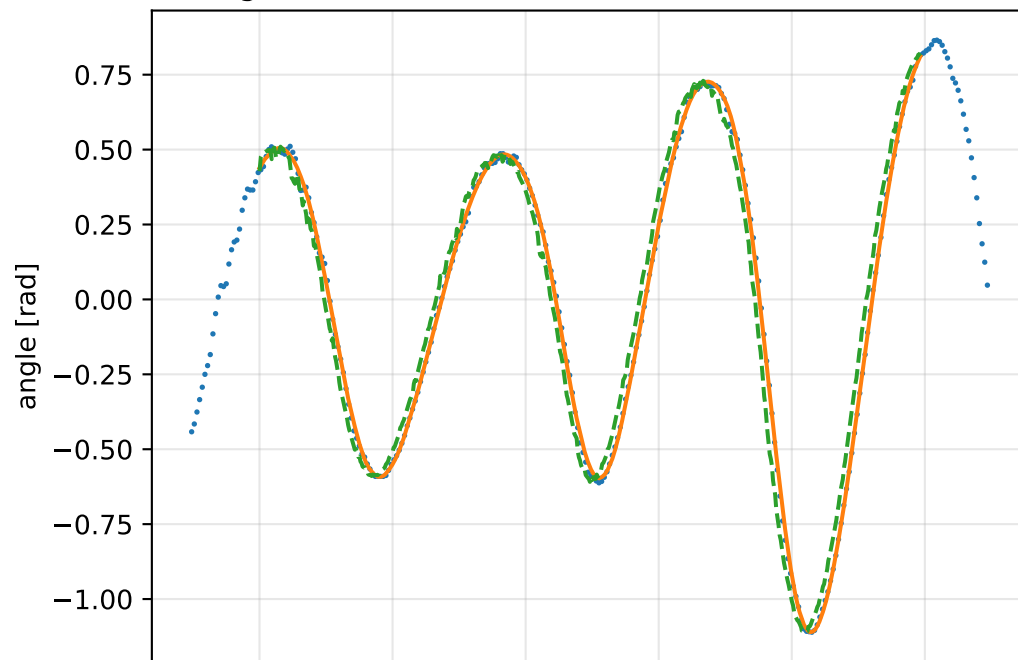


lag_legacy_smooth_schedule: gimbal measurement audit (objective=measured_sg)

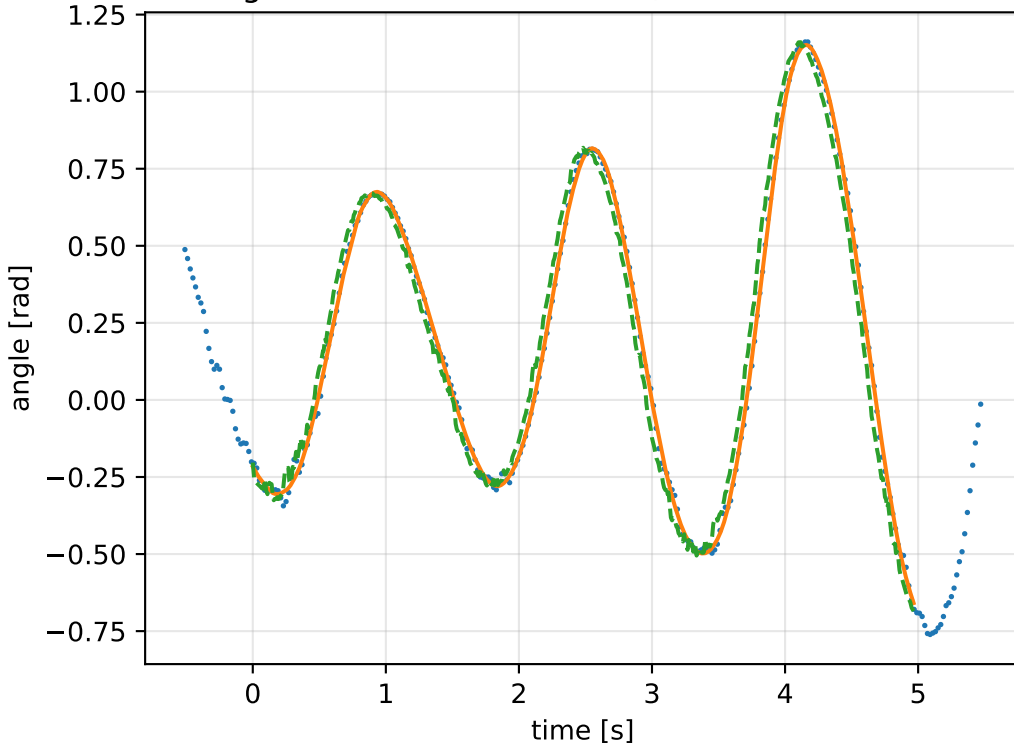
gimbal 1: RMSE=0.0759, max=0.2095 rad



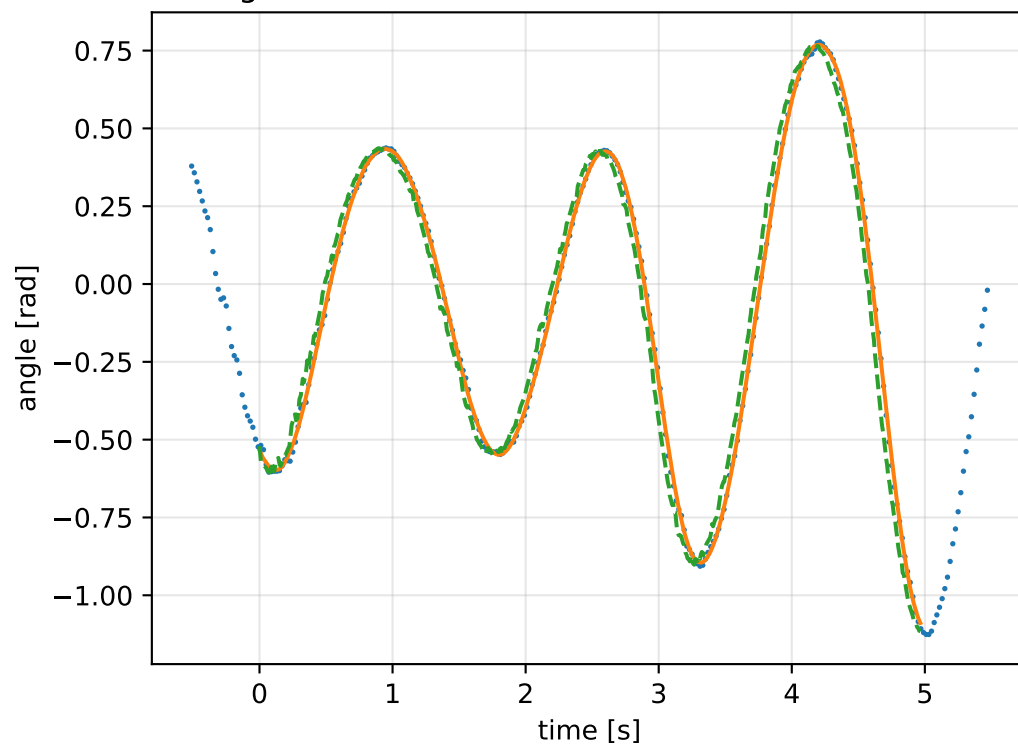
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

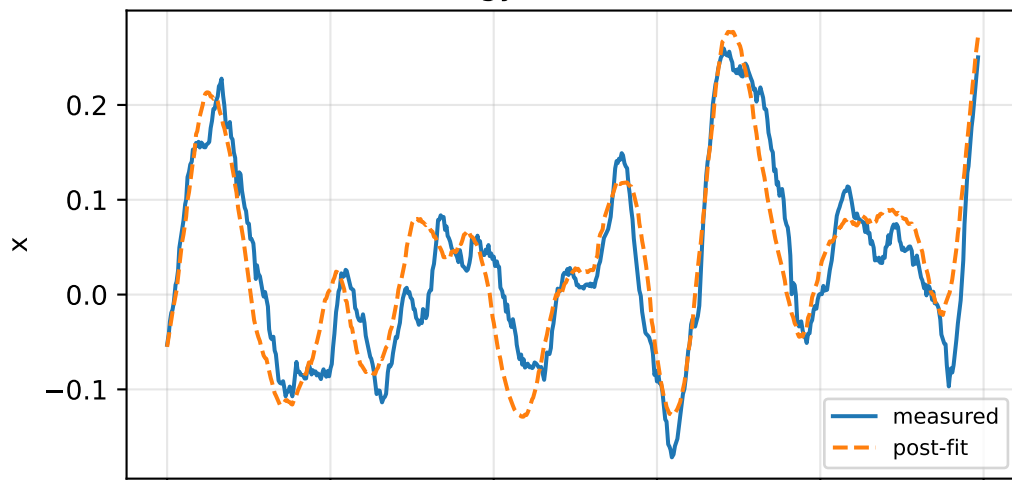


gimbal 4: RMSE=0.07131, max=0.1737 rad

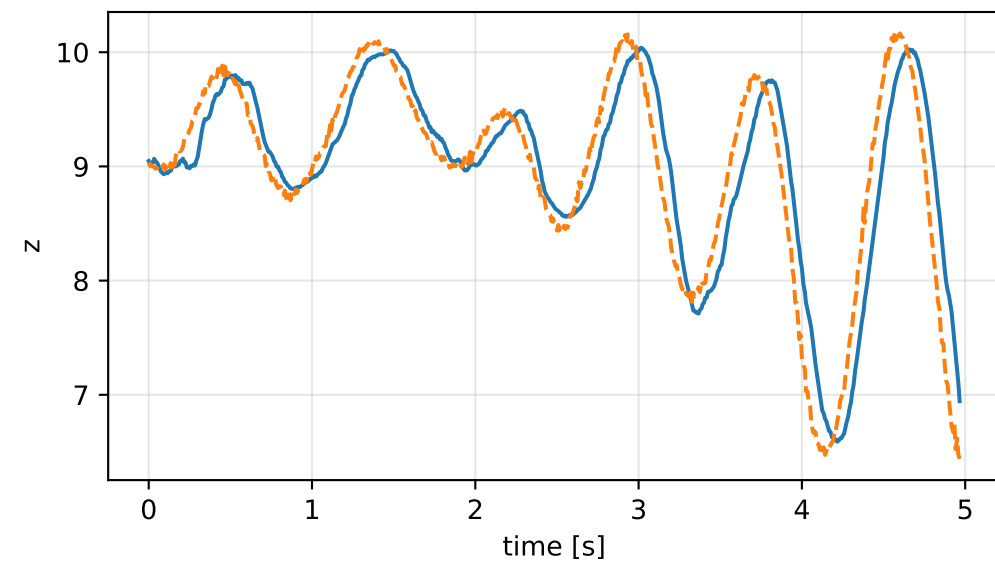
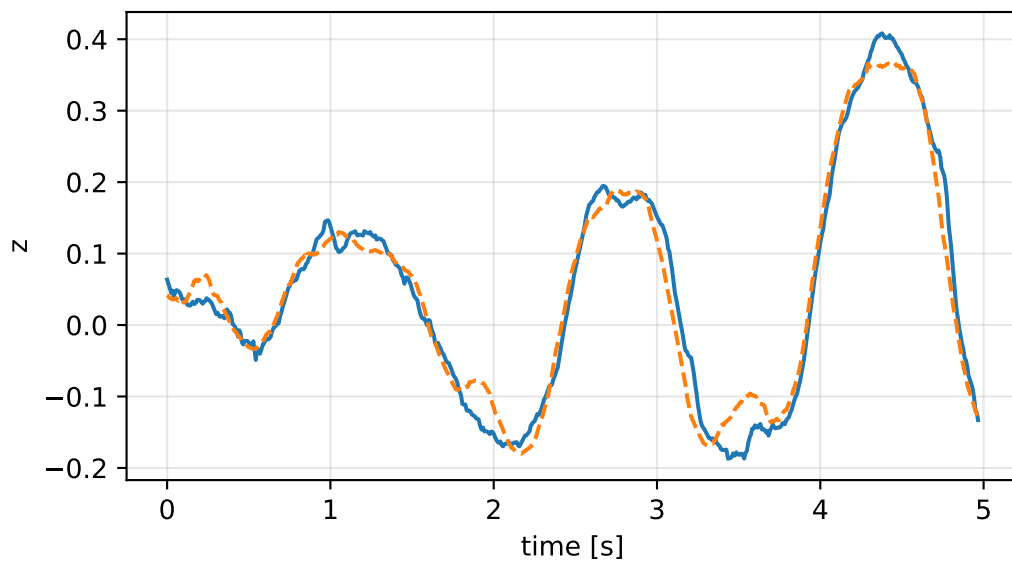
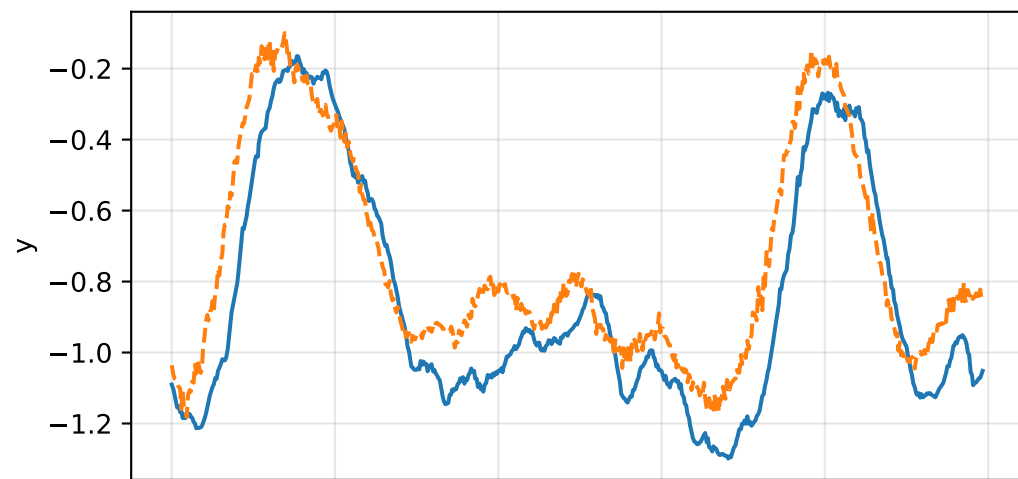
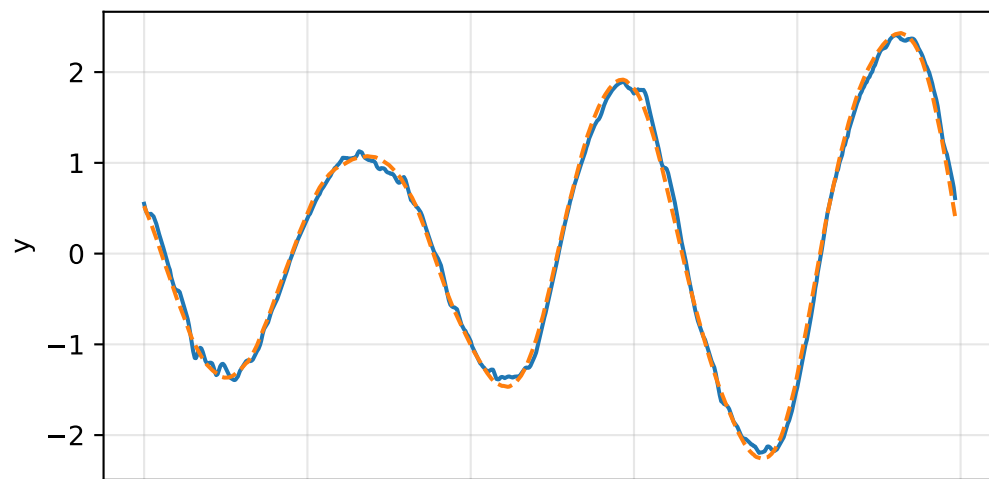
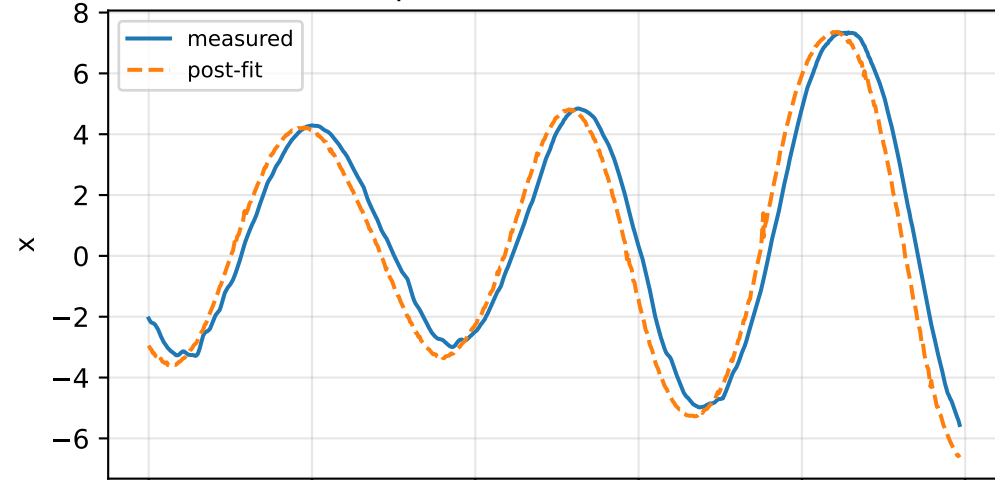


lag_legacy_smooth_schedule: IMU comparison (diagnostic only)

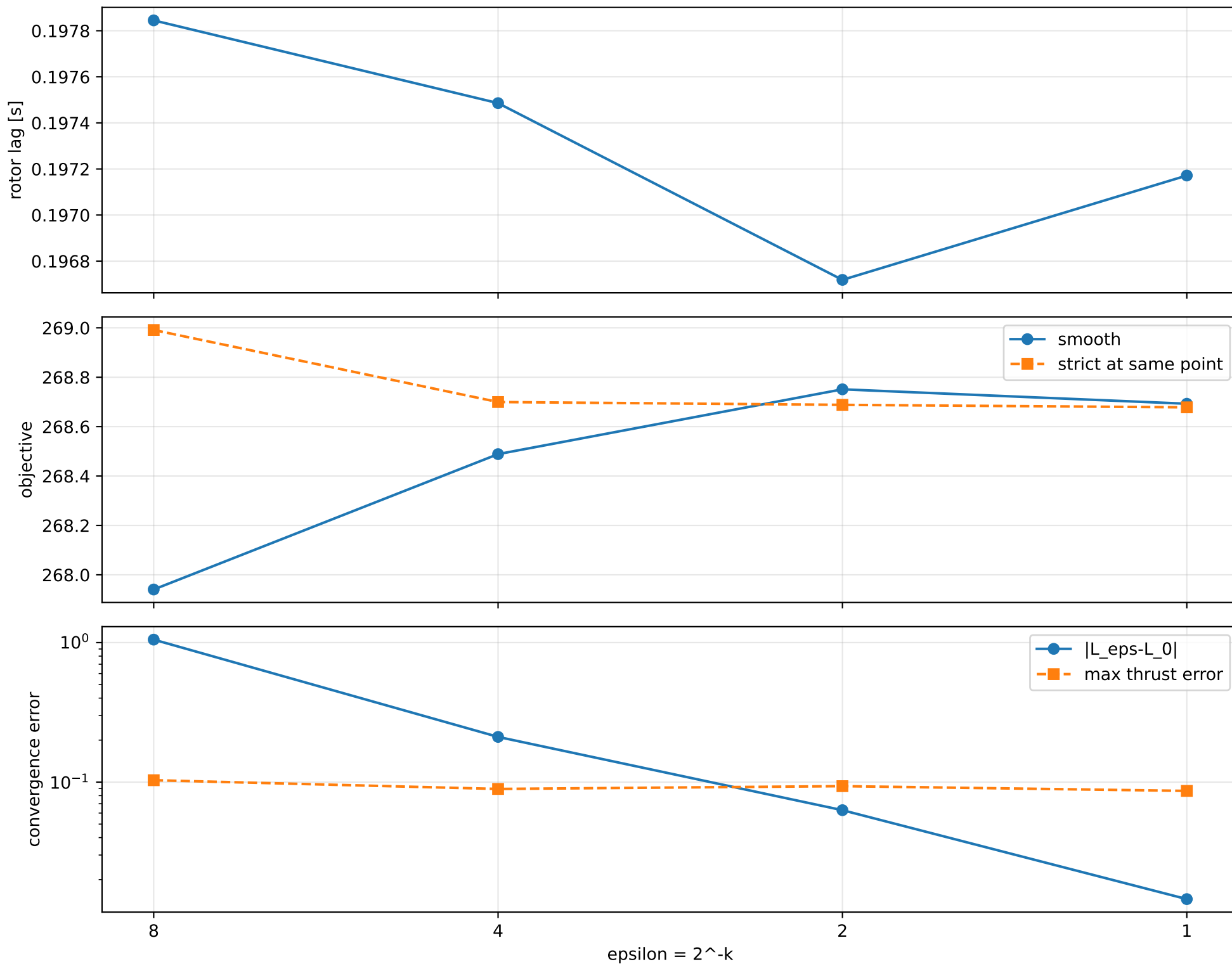
gyro [rad/s]



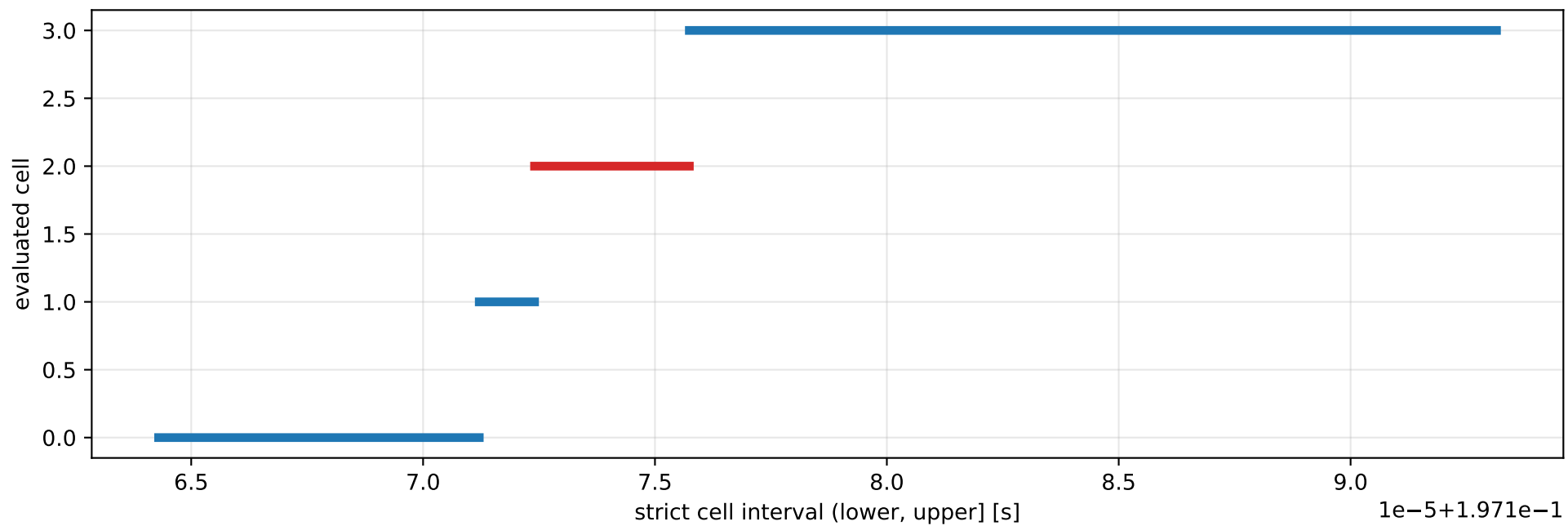
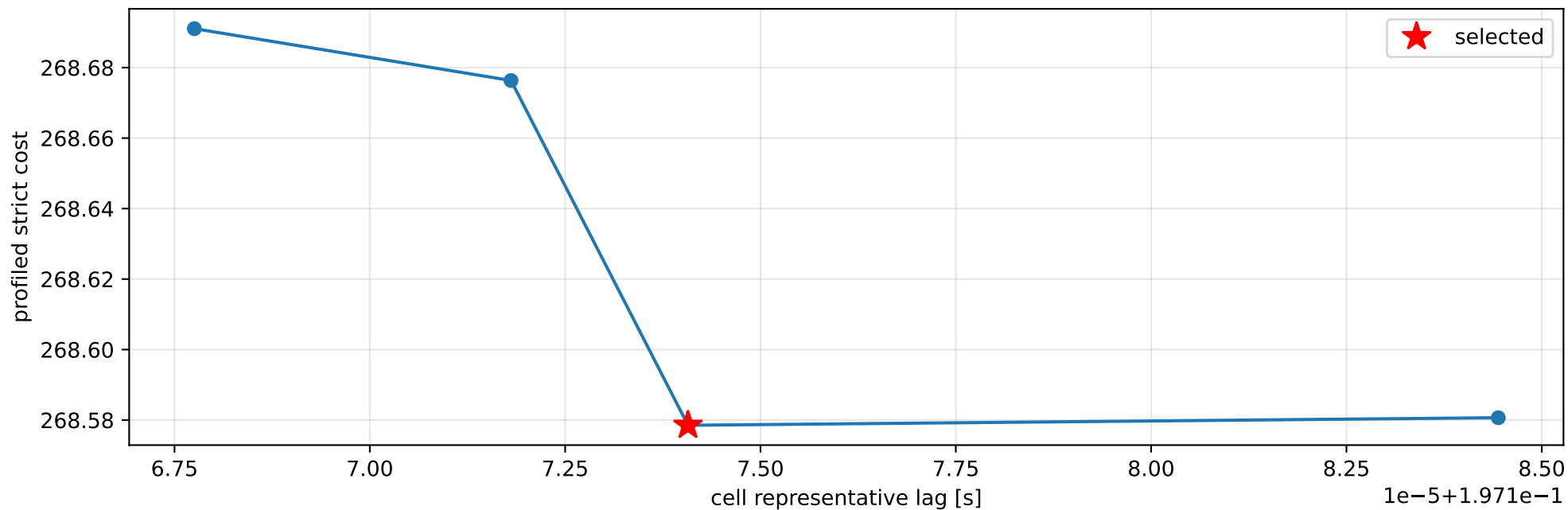
specific force [m/s²]



lag_legacy_smooth_schedule: smooth-to-strict continuation



lag_legacy_smooth_schedule: exact strict-ZOH lag cells



selected (0.197172403, 0.197175741] s; final smooth 0.197171549 s; data boundary=False

lag_legacy_smooth_schedule: parameter estimate

Case: lag_legacy_smooth_schedule

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.24407559

inertia [kg m²]:

[[4.31176004e-01 -5.24018991e-04 -7.00307487e-02]

[-5.24018991e-04 3.28436448e-01 4.56754859e-05]

[-7.00307487e-02 4.56754859e-05 7.28247145e-01]]

CoG body [m]: [0.01763034 -0.00412155 -0.04442266]

force effectiveness: [1.42142877 1.83042594 1.59832228 1.32510768]

scale-free J/m [m²]:

[[1.01594798e-01 -1.23470702e-04 -1.65008250e-02]

[-1.23470702e-04 7.73870401e-02 1.07621754e-05]

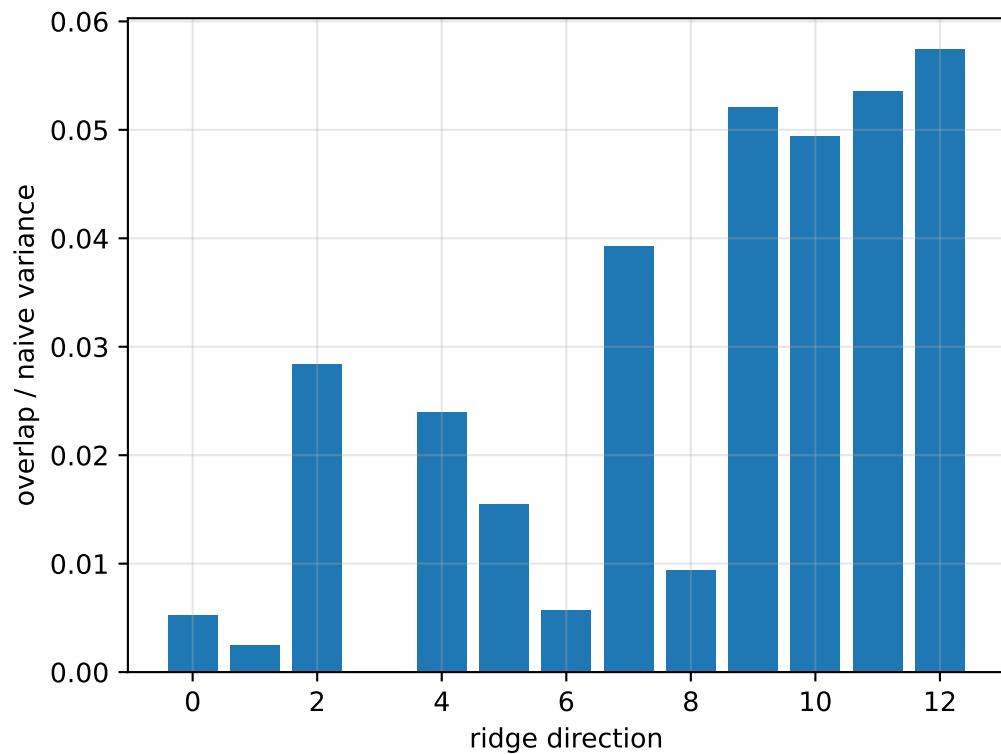
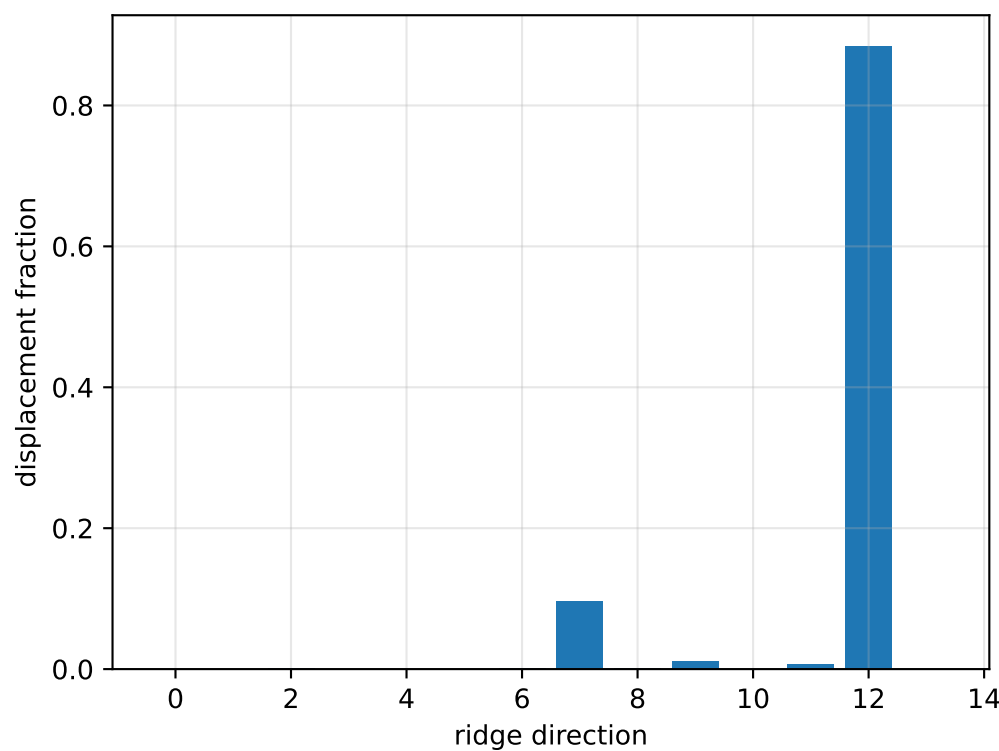
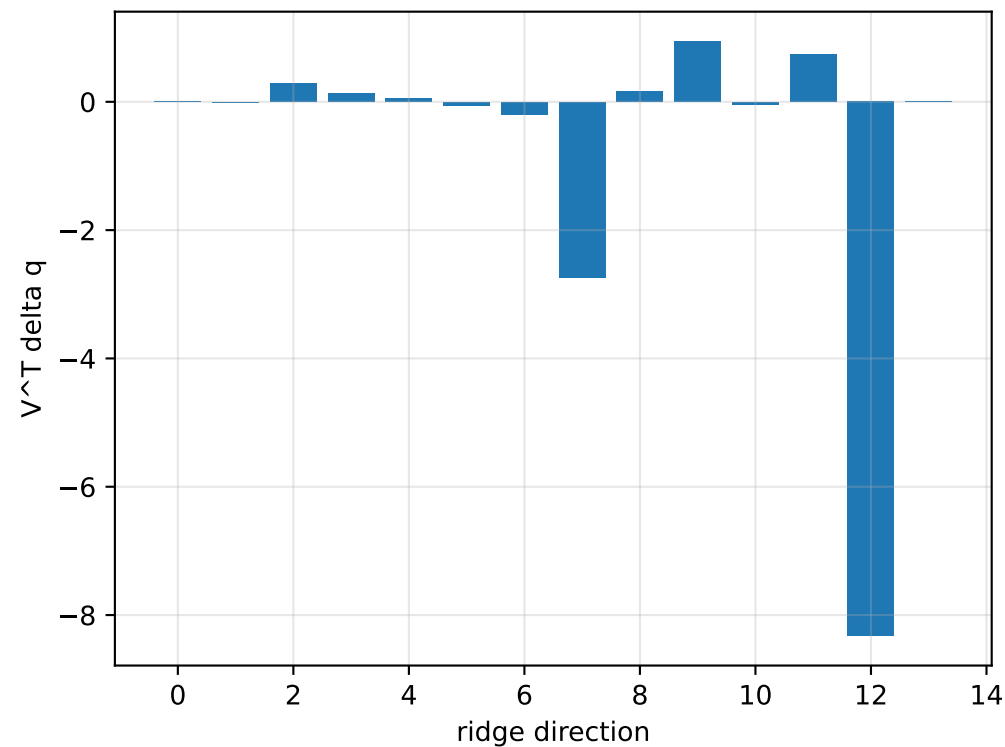
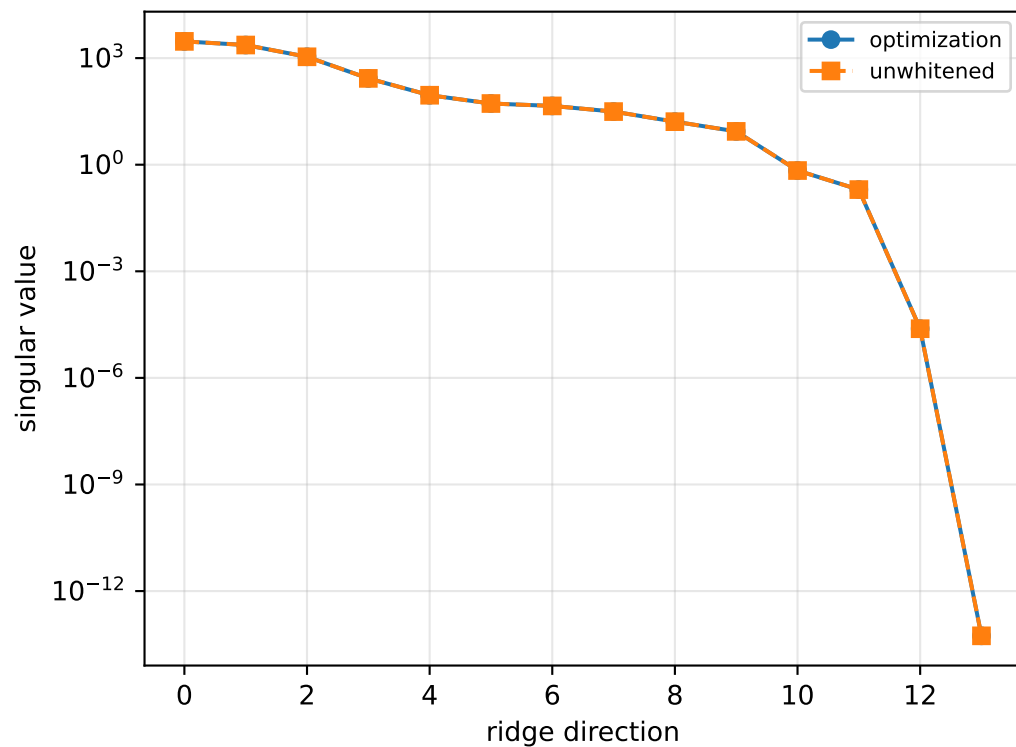
[-1.65008250e-02 1.07621754e-05 1.71591464e-01]]

scale-free f/m: [0.3349207 0.43128967 0.37660093 0.31222528]

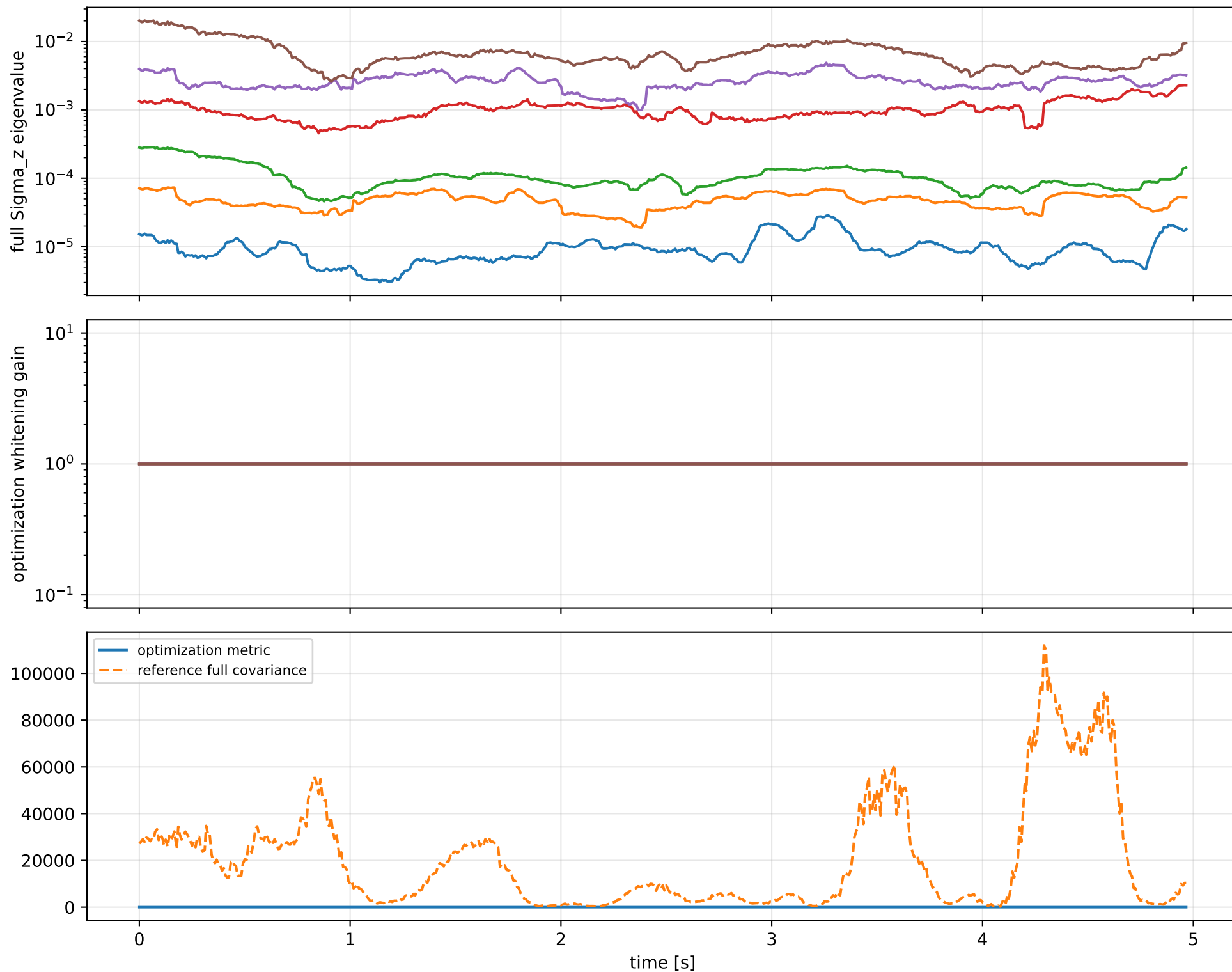
rotor lag cell: (0.197172403, 0.197175741] s

representative: 0.197174072 s

lag_legacy_smooth_schedule: ridge spectrum (rank 13, nullity 1, ||Jv||=1.77e-13)



lag_legacy_smooth_schedule: optimization vs reference covariance



lag_legacy_smooth_schedule: wrench / closure (force max 1.21e-14, torque max 1.26e-15)

